



LlamaTrade

PRODUCT ROADMAP

MVP Release Plan

From build to launch

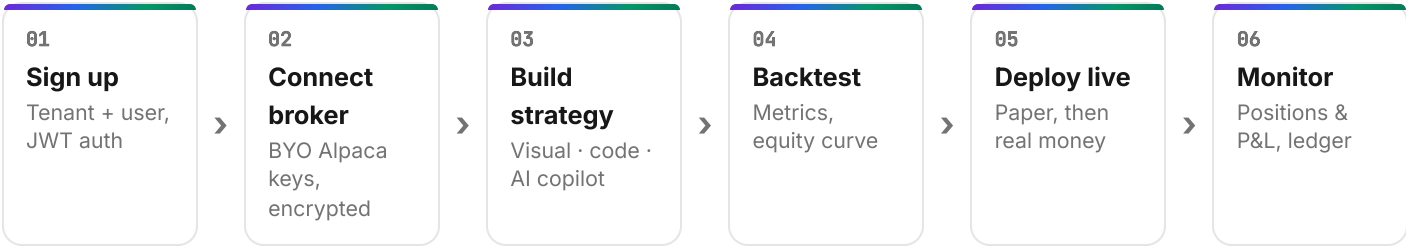
A milestone-by-milestone path from today's implementation to a shippable MVP — framed around what each stage unlocks for testing, and who gets to use it. Built for the product team to plan beta cohorts and validation.

STATUS	DATE	MVP SHAPE	OWNER
DRAFT FOR REVIEW	June 16, 2026	Invite-only paper beta	Product & Engineering

01 What LlamaTrade is

A multi-tenant SaaS where traders **visually build algorithmic strategies, backtest them against historical data, and run them live** against their own Alpaca brokerage account — without LlamaTrade ever holding their money. Users bring their own Alpaca keys; we orchestrate strategy logic, execution, and accounting on top.

The differentiator is the **visual, no-code strategy builder** backed by a real DSL and an **AI copilot**, paired with a **double-entry ledger** that lets multiple strategies share one brokerage account without colliding on capital — something most retail algo platforms do poorly.



02 Where we are today

The hard parts are built and tested. The strategy engine, backtest engine, live-execution backend, and the double-entry portfolio ledger are real and covered by tests. The MVP gap is mostly **UI surface, security hardening, and onboarding polish** — not core trading logic.

CAPABILITY	MATURITY	NOTES
Build strategy (visual + code)	Ready 80%	Block builder + S-expression DSL + validation. 16+ indicators, multi-symbol, live=backtest parity.
Backtest end-to-end	Ready 80%	Celery exec, full metrics, benchmark, live progress streaming, cooperative cancel.
Market data	Ready 95%	TimescaleDB store + Alpaca + streaming + resilient client.
Live / paper execution (backend)	Ready 85%	All order types, fills, crash-safe order IDs, risk checks. No UI yet.
Portfolio ledger	Ready 80%	Double-entry, sleeves, FIFO, reconciliation. Read cutover behind a flag, pending soak.

CAPABILITY	MATURITY	NOTES
Connect Alpaca (BYO keys)	Backend 85%	Encrypted creds + per-session clients fully built — no UI exists to enter/manage keys.
Sign up / login / tenancy	Partial 60%	JWT + bcrypt + isolation work. No email verify / password reset / token revoke.
AI copilot (Agent)	Partial 45%	LLM chat + DSL/backtest tools. Not wired into the builder UI.
Billing (Stripe)	Partial 55%	Subs/payments/webhooks work; limits not enforced; no UI gating. Not on MVP critical path.
Trading page (UI)	Stub ● 10%	Placeholder only — the biggest single product gap.
Infrastructure / security	Hardening 50%	Docker/K8s/Terraform/CI solid. Gaps: TLS, inter-service auth, secrets-in-git, deploy automation, trace/log shipping.

03 Locked scope decisions

These were decided with the product owner and shape the entire plan:

DECISION	CHOICE	CONSEQUENCE
Trading scope at launch	Paper-first · live as fast-follow	MVP carries zero real-money risk. Real money ships after the beta proves the loop, gated on extra hardening + legal disclosures.
Launch audience	Closed / invite-only · beta	Self-serve billing & email verification drop out of the MVP gate — invited testers are provisioned manually. Hardening only needs "safe for invited externals."
Non-core surfaces	AI copilot IN · others deferred	The Agent copilot is part of the MVP differentiator. Notifications, manual trading, and corporate actions are deferred.
Brokerage model	Bring-your-own Alpaca keys	We automate the user's <i>own</i> account, never custody funds. Trading customers' money (Broker API, KYC, RIA posture) is a separate legally-gated program.

04 The MVP, defined

MVP DEFINITION

A closed, invite-only, paper-trading beta in which an invited user can: **sign up** → **connect their Alpaca paper account** → **build a strategy** (visually, in code, or via the AI copilot) → **backtest it** → **deploy it in paper mode** → **and watch orders, positions, and P&L update live** — on a deployment that's safe to expose to invited external testers.

The MVP is reached after milestones **M1 + M2 + M3 + M4**. M5 and M6 are post-MVP fast-follows toward public self-serve and real money.

Effort legend (single engineer, calibrate with your team): **S** ≈ a few days · **M** ≈ 1–2 weeks · **L** ≈ 2–4 weeks. Relative sizes, not a committed calendar.

05 Milestones to MVP

M1 Build & Backtest

INTERNAL ALPHA

Unlocks: a user signs up, builds a strategy, runs a backtest, and sees real metrics & equity curves. First validation of the core value prop — builder UX and backtest output — with **zero brokerage involvement**.

Audience: internal team + friendly users on staging.

SCOPE

- Verify / repair backtest **results data-binding** in the UI (trades, monthly returns, metrics). [M]
- Build a real **Dashboard** — snapshot, recent backtests, empty-states. [M]
- Remove the **silent demo-data fallback** in Portfolio so failures surface honestly. [S]
- Seed **3–5 strategy templates** so the builder isn't a blank page. [S]

ACCEPTANCE

A user with no prior data can register, open a template, edit it, save, run a backtest, and read **correct** results — with no mock data anywhere in the path.

DEPENDENCIES

None. ~90% already works — fastest path to real feedback.

M2 Paper Trading Live

CLOSED BETA CORE

Unlocks: an invited user connects an Alpaca **paper** account, funds a strategy, deploys it, and watches orders fill and positions/P&L update in real time. The first end-to-end test of the **entire live loop** (execution → fills → ledger → portfolio) with no real money. The heart of the beta.

SCOPE

- **Broker credential UI** + Phase-0 security fixes: make secrets write-only, validate keys against Alpaca on entry. [M]
- **Trading UI built from scratch:** session start/stop, credential + strategy picker, live order/position/fill stream, sleeve funding. [L]
- Wire frontend to `StartSession` + **Redis-Streams** live updates. [M]
- Complete the `LEDGER_READS` soak; cut portfolio reads onto the ledger. [M]
- Onboarding disclosures (history begins at onboarding; metrics need ~30 days). [S]

ACCEPTANCE

An invited user connects a paper account, deploys a strategy, sees a real order fill in the UI within seconds, and sees position + P&L in Portfolio — ledger reconciliation green against the broker.

DEPENDENCIES

M1, plus broker-setup Phase 0 (security) as a hard gate.

M3 AI Copilot in the Builder

BETA DIFFERENTIATOR

Unlocks: a user describes a strategy in plain language and the copilot generates/edits the DSL, validates it, and can run a backtest — inside the builder. Tests whether conversational strategy creation lowers the build barrier (a key product bet). Runs in parallel with M2.

SCOPE

- Wire the existing **Agent service** into the builder UI: chat panel, streaming, apply generated DSL into the editor. [M]
- Close the Agent's **tool-result feedback loop** (tools execute but don't fully feed back today). [M]
- Guardrail: every copilot strategy passes the DSL validator before save/deploy. [S]

ACCEPTANCE

"Build me a momentum rotation across tech ETFs, rebalance monthly" yields a valid strategy in the builder, backtestable without hand-editing.

DEPENDENCIES

M1 (builder + backtest). Parallel to M2.

M4 Beta Hardening

SAFE TO INVITE EXTERNALS

Unlocks: the platform can go in front of non-employee invited users without a security incident. Not user-visible, but it's the gate that turns M1-M3 from "demo on staging" into "a beta people can log into."

Independent — can start now; parallel to M1-M3.

SCOPE (SCALED TO INVITED BETA)

- **TLS** on ingress (provision the missing managed certificate). [S]
- **Inter-service auth** — add bearer tokens (lighter than full mTLS for beta). [M]
- **Secrets out of git** — External Secrets Operator / Secret Manager. [M]
- **Credential encryption** — per-value salts + rotation (KMS deferred to M6). [M]
- **Deploy automation** — roll all 9 services + await migrations (today only auth + frontend). [S]
- **Observability** — ship traces to Cloud Trace, logs to Cloud Logging; finish SLO alerts. [S]
- Add the **Agent service K8s manifests** (missing today). [S]

Acceptance: staging reachable over HTTPS, no secrets in git, services authenticate to each other, one command rolls the stack after migrations, tester traffic visible in traces/logs.

◆ MVP GATE ◆

Closed Invite-Only Paper Beta

Validates the build → backtest → paper-trade → monitor loop, the AI-copilot build path, ledger correctness against a live broker, and onboarding friction — with real users and zero financial risk.

06 Post-MVP fast-follows

M5 Self-Serve & Billing

PUBLIC SIGNUP

Unlocks: strangers can sign up, verify email, subscribe with a card, and have plan limits enforced — turning the beta into a public product.

SCOPE

- Email verification + password reset. [M]
- Auto-create Free subscription on signup. [S]
- Enforce plan entitlements server-side **and** reflect as locked/disabled UI. [M]
- Usage metering for backtest/strategy limits. [M]
- Basic dunning on failed payments. [S]

Dependencies: MVP gate. Billing backend is ~55% done already.

M6 Real-Money Live Trading

GENERAL AVAILABILITY

Unlocks: users flip from paper to **real-money** trading — the full product.

SCOPE

- Enable live mode behind paid-plan gating + explicit confirmation/guardrails. [S]
- Upgrade credential encryption to **KMS envelope** encryption. [M]
- Load-test reconciliation + ledger at realistic account counts. [M]
- Failover / DR test — Cloud SQL HA, backup restore. [M]
- Support runbooks + on-call dashboards + SLO burn-rate alerts. [M]
- **Risk/legal disclosures** for live trading — needs product + legal sign-off. [S]

Long pole: the legal disclosure work — start it early. Trading *customers'* money is a separate program, not M6.

07 Sequencing & parallelization

CORE PATH

M1 · Build & Backtest

M2 · Paper Trading Live

— MVP gate —

M5 · Self-Serve

M6 · Real Money

PARALLEL

M3 · AI Copilot

INDEPENDENT

M4 · Beta Hardening — start immediately, must finish before the beta opens

M1 is the unblocker — fastest to real feedback (~90% built). **M2, M3, M4 run in parallel** after M1 (M4 can start today). M2 is the largest (UI-bound); M3 and M4 are medium. **M5 and M6 do not block the MVP** and should be scoped once beta feedback is in. Rough weight to the gate: **M2 (L) > M4 (M-L) ≈ M3 (M) > M1 (M)**.

Risks & open decisions

RISKS

Trading UI is greenfield (M2)

Backend is ready, but the entire live surface is new — the schedule's long pole to the gate.

Ledger read soak must prove parity

`LEDGER_READS` needs to soak in staging before M2 can trust portfolio numbers. Budget time for the soak, not just the code.

Copilot quality bar (M3)

LLM-generated strategies must always pass the DSL validator; a bad generation that silently saves is a trust killer.

Docs lag code

Several planning docs describe services as more stubbed than they are (backtest was rebuilt). Treat code as source of truth.

OPEN DECISIONS — NEED A PRODUCT/ENG CALL

Broker connect UX

Manual API-key paste vs. Alpaca OAuth (no raw secret stored, natural bridge to advisor-on-client later).

Recommend manual keys for v1, OAuth as a fast-follow.

Credential encryption bar for beta

App-layer hardened (salt + rotation) vs. KMS envelope encryption.

Salt+rotation is enough for paper beta; KMS required before M6 (real money).

Multi-user tenants in beta?

Credentials are tenant-scoped with no per-user ACL or audit today.

Fine for single-operator beta; decide before multi-seat firms.

Notifications in beta?

Currently deferred. If testers need fill/drift alerts, email-only delivery is the minimum add.

Defer unless testers explicitly need it.

Source & detail: the engineering breakdown lives in `.docs/planning/mvp-release-plan.md` and links to the broker-setup, ledger, Redis-Streams, and telemetry plans. This deck is the product-facing summary of those.